

Team Reflection

Title :- Rocket Landing And Docking

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Category: B

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Q1: What was the single hardest technical challenge you faced in deriving the mathematical model? How did you resolve it?

The hardest challenge was deriving an accurate dynamic model for the drone that included rotational motion (pitch, roll, yaw) and motor thrust relationships. Initially, our equations were too simplified and did not account for coupling between axes, which caused inconsistencies in simulation results. We resolved this by revisiting Newton-Euler equations and breaking the system into smaller parts (forces, torques, inertia). We also cross-checked our derivation with reference models and validated each step through simulation.

Q2: Which Course Outcome (CO1-CO5) was most challenging for your team? Why?

CO3 (applying mathematical techniques to real-world systems) was the most challenging. While we understood the theory, applying it to a practical system required assumptions, approximations, and interpretation of physical meaning. Translating equations into a working model exposed gaps in our conceptual clarity.

Q3: What would you do differently if you started the project over?

We would begin with a clearer plan for modeling and validation instead of directly jumping into derivations. Creating a block diagram early on and identifying all assumptions beforehand would have saved time. We would also start simulations earlier to catch errors sooner.

Q4: What did you learn from this project that a regular textbook problem could NOT teach you?

This project taught us how messy real-world modeling is compared to textbook problems. In textbooks, all variables are defined and equations are clean, but in reality, we had to decide what to include, what to ignore, and how to validate results. We also learned how small errors in assumptions can significantly affect the final outcome.

Q5: How did you divide work among team members? Did everyone understand the full derivation?

We divided the work based on strengths: two members handled derivations, another two worked on simulations, and another focused on documentation. However, we conducted regular discussions to ensure everyone understood the complete model. By the end, all members had a basic understanding of the full derivation, though some were more confident in specific parts.

Q6: What AI tools did you use? For what specific purpose? What did you NOT use AI for?

We used AI tools like ChatGPT to clarify concepts, verify intermediate steps in derivations, and improve the clarity of our report. We also used it to debug equations and understand theoretical gaps. However, we did not rely on AI to generate the full solution or derivation; all key steps were done and verified by our team to ensure correctness and understanding.